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EXPERIENCE

Hashed In by Deloitte

Software Engineer 1

On-Site - Chennai, India

Nov 2025 – Present

Autonomous Coding

- Architected an **Agentic AI** system using Django to automate the generation of XML, XSLT, SQL, and Oracle Fast Formulae, streamlining internal development workflows, leveraging Azure Cosmos DB for DBI retrieval.
- Integrated **Azure OpenAI** and **Anthropic models** to implement rule-based flow control and a company knowledge-base for contextual data validation.
- Engineered end-end generation of Oracle eText and RTF templates, incorporating automated conditional formatting and logic validation using AI models.
- Leveraged **RAG** (Retrieval-Augmented Generation) principles to provide company-specific Q&A and contextual validation within the automation pipeline.

Multi-Agent Code Analysis System

- Architected a **Multi-Agent** System using **LangGraph** to automate codebase analysis, generating role-specific documentation for Software Engineers (technical) and Product Managers (business-logic).
- Developed an Interactive Agentic Pipeline with **FastAPI** that supports real-time state management, allowing users to inject custom instructions to steer the analysis.
- Engineered Semantic Code Retrieval using **pgvector** within a **PostgreSQL** (pgAdmin) environment to chunk, store, and query code metadata for high-context agent responses.
- Integrated Agents using **LangChain** and **RAG** principles to search for external documentation and best practices, ensuring the system provides up-to-date analysis of frameworks and libraries.
- Built a Real-Time Monitoring Dashboard in **Streamlit** that visualizes agent progress, provides a project-based user interface, and enables natural language Q&A over the analysed codebase.

Microchip Technology

Engineer I - Corporate Applications Engineer

On-Site - Chennai, India

Jul 2024 – Mar 2025

- Worked with clients to deliver critical project tasks across the entire WNET (L1 and L2) product portfolio, consistently meeting deadlines.
- Developed a **Python-based tool using OpenCV** to extract pin names from customer schematics, significantly accelerating design validation and enabling usage across global engineering teams.
- Designed and implemented a **custom network testing tool in Python**, tailored for proprietary hardware platforms. The tool improved test efficiency and reduced manual effort by **30%**.
- Developed a **multi-screen GUI application (6 screens)** with structured modular design, and packaged it into a standalone executable. Conducted comprehensive testing to ensure seamless functionality and deployment.

Student Intern

Microchip Technology

Chennai, India

Jan 2024 - Jun 2024

- Designed and implemented an embedded C solution to enable Ethernet communication on entry-level microcontrollers lacking native Ethernet support, achieving a **25% improvement in data transmission speed** through software optimization.
- Developed a **lightweight software utility** for an **hourly reminder system** with configurable intervals, optimized for low-power operation on resource-constrained embedded platforms, ensuring efficient performance and extended runtime.

Workshop

Bangalore, India (Sep 2022)

- Gained hands-on experience with Android development tools and implemented real-time text-to-speech functionality conducted by **IISc Bangalore** to enhance practical knowledge in mobile app development.

EDUCATION

Thiagarajar College of Engineering

Bachelors of Engineering in Electronics and Communication Engineering

Madurai, India

Sep 2020 – May 2024

CGPA 9.61

Vidhya Sagar Global school

12th Grade

Chengalpattu, India

June 2019 - March 2020

Score 92%

PROJECTS

- **Email Automation Script:** Developed a **Python** script using **Selenium WebDriver** to automate email sending for a college club event, successfully delivering over **500 invitation emails** to participants.
- **Car Showroom Management System:** Developed a desktop-based application with full **CRUD** functionality to manage vehicle inventory, track status updates, and streamline overall showroom operations.
- **Hostel Management System:** Built a web-based student record management system with complete **CRUD** operations, integrating a relational database for efficient data storage, retrieval, and management.
- **Smog Prediction and alerting system:** Executed **end-to-end integration** of a smog detection system using NodeMCU, environmental sensors, and custom PCB for interfacing. Enabled real-time and cloud-based monitoring with ThingSpeak, applying **data analytics** and **predictive modeling (LPC)** for trend analysis. Combined hardware design, real-time data processing, and IoT for smart environmental monitoring.
- **Hobby Projects:**
 - Designed and implemented game logic for Tic-Tac-Toe and Black Jack using **Python**.
 - Implemented web scraping projects using **Python**, leveraging libraries like Requests, BeautifulSoup, and CSV for data extraction and organization.

SKILLS

- **Languages:** Python, PHP, JavaScript, SQL, C
- **Frameworks :** FastAPI, Django, Streamlit
- **Database:** PostgreSQL, Azure CosmosDB
- **Generative AI :** Langchain, Langgraph, Autogen, OpenAI, Anthropic
- **Version Control:** Git
- **Web Technologies:** HTML, CSS

CERTIFICATIONS

- Building AI Agents & Agentic AI system via Microsoft Autogen - Udemy
- FastAPI – The complete course (Beginner + Advanced) - Udemy
- Elite certification for “The Joy of Computing using Python” awarded by IIT-Madras
- The Complete Python Bootcamp from Zero to hero in Python - Udemy
- The complete 2024 Web Development Bootcamp - Udemy.